

- 6 (a) A length of copper wire of cross-sectional area $1.2 \times 10^{-6} \text{ m}^2$ carries a steady current of 2.5 A. The wire has a density of $8.9 \times 10^3 \text{ kg m}^{-3}$. Assume the wire is made entirely of copper atoms, each contributing one free electron to conduction. The molar mass of copper is 63.5 g mol^{-1} .

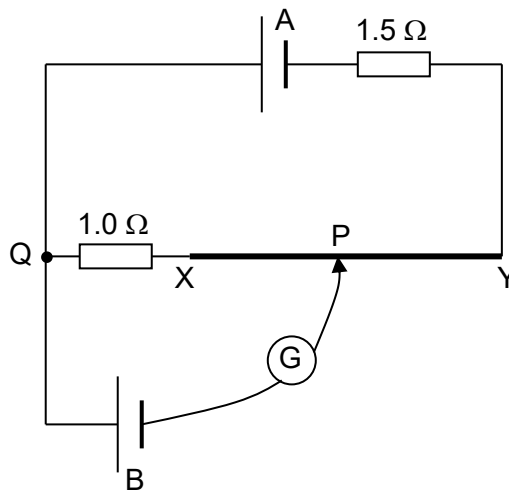
Calculate the average drift velocity of the electrons in the wire.

$$\begin{aligned} \text{number density } n &= \frac{\text{No. of atoms}}{\text{Volume}} = \frac{\text{mass}}{(\text{molar mass})(\text{Volume})} \times N_A \\ &= \frac{\text{density}}{\text{molar mass}} \times N_A \\ &= \frac{8.9 \times 10^3}{0.0635} \times 6.02 \times 10^{23} \quad [1] \end{aligned}$$

$$\begin{aligned} \text{Drift velocity } v &= \frac{I}{neA} \quad [1] \\ &= \frac{(2.5)(0.0635)}{(8.9 \times 10^3)(6.02 \times 10^{23})(1.6 \times 10^{-19})(1.2 \times 10^{-6})} \\ &= 1.5 \times 10^{-4} \text{ m s}^{-1} \quad [1] \end{aligned}$$

average drift velocity = m s^{-1} [3]

- (b) In the circuit below, cell A has an e.m.f. 2.0 V and negligible internal resistance. Wire XY is 100.0 cm long with a resistance of 5.0Ω .



- (i) Distinguish between *electromotive force e.m.f.* and *potential difference p.d.* using energy considerations.

E.m.f. is the electrical energy converted from other forms of energy per unit charge delivered by the source. *P.d.* is the electrical energy converted to other forms of energy per unit charge when charges pass from one point to another. [1]

- (ii) Calculate the current flowing from Q to Y when the galvanometer registers a null deflection.

When the galvanometer registers a null deflection, no current in the lower circuit.

$$I = \frac{2.0}{1.0 + 5.0 + 1.5} \quad [1]$$

$$= 0.27 \text{ A} \quad [1]$$

current = A [2]

- (iii) Cell B has an e.m.f. of 1.5 V. At balance point P,

1. show that resistance across X and P is 4.6 Ω ,

At balance point, $V_{QP} = \mathcal{E}_B = 1.5 \text{ V}$ [1]

By the potential divider principle,

$$V_{QP} = \frac{1.0 + R_{XP}}{R_{total}} \times \mathcal{E}_A$$

$$1.5 = \frac{1.0 + R_{XP}}{1.0 + 5.0 + 1.5} \times 2.0 \quad [1]$$

$$R_{XP} = 4.625 \Omega = 4.6 \Omega \quad (2 \text{ s.f.})$$

[2]

2. calculate the balance length XP.

$$\frac{L_{XP}}{L_{XY}} = \frac{R_{XP}}{R_{XY}} \quad [1]$$

$$L_{XP} = \frac{4.625}{5.0} \times 1.0 = 0.925 \text{ m} \quad [1]$$

length XP = m [2]

- (iv) State and explain how the length XP in (b)(iii)2. will change if the internal resistance of cell A is not negligible.

From potential divider principle, the p.d. across the wire XY will now be smaller. [1]
Hence length XP will be longer. [1]

.....[2]

[Total: 12]

